

PROFILE

Software Engineer with over 3 years of experience specializing in Python and Java backend development, with hands-on work in Generative AI, Retrieval-Augmented Generation (RAG) systems, and vector databases. Skilled in API design, data parsing, microservice architectures, and shipping inside large enterprise codebases with significant legacy and migration work.

PROFESSIONAL EXPERIENCE

Cognizant

Pune, India

Associate, QEA Tech COE - Generative AI

Nov 2023 – Present

- Core R&D Engineer in the Machine Learning and Generative AI team at the Cognizant QEA TechCoE, contributing to AI-driven products that enhance Cognizant's QEA offerings.

Generative AI Assurance Platform · Jan 2026 – Present

- Migrating a production Flask backend to FastAPI while extending the API surface for new product features.
- Remediating data quality issues in the underlying database to support the modernized service layer.
- Coordinating with frontend developers to align the new React UI with revised backend contracts.

Cognizant

Pune, India

Programmer Analyst, QEA Tech COE - Marketplace

Aug 2022 – Dec 2025

AILA — React Application Modernization · Jul 2025 – Nov 2025

- Led component-level migration of a legacy React codebase to modern React, working through a prioritized component matrix.
- Remediated security vulnerabilities flagged during IP-readiness review and shipped the modernized application to delivery.

Knowledge Management Chatbot — RAG Platform · Dec 2023 – Oct 2024

- Designed and implemented microservice APIs for the LLM Chatbot backend using FastAPI, accelerating development time by 35%.
- Developed an advanced Python parser extracting data from complex PowerPoint slides, increasing text retention by 30% and improving knowledge base quality.
- Integrated Qdrant Vector Database, boosting search response accuracy by 40% and reducing retrieval time by 50%.
- Applied caching mechanisms reducing search latency by 60%; secured admin API endpoints with OAuth2 authentication.

Document vs Webpage Content Comparator · Oct 2022 – Nov 2023

- Contributed to the development of the QEA TechCoE Marketplace Platform, enhancing automation and efficiency in software engineering processes.
- Owned and developed the **Document and Webpage Comparator Bot**, optimizing backend logic in Java with Apache PDFBox and reducing processing time by 25%.
- Designed a heuristic paragraph-segmentation algorithm based on statistical line-length analysis to recover semantic blocks from raw extracted text.
- Applied NLP algorithms to detect mismatches in word-, sentence-, and sequence-level alignment between PDFs and websites, increasing accuracy by 45%.

Cognizant

Pune, India

Programmer Analyst Trainee

Feb 2022 – Jul 2022

- Implemented back-end logic in Core Java with robust error handling and OOP principles; developed responsive web pages in HTML, CSS, and JavaScript and debugged complex production scenarios to reduce issue resolution time by 25%.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, CSS**Frameworks:** FastAPI, Spring Boot**AI / ML:** Retrieval-Augmented Generation (RAG), sentence-transformers, LangChain, embedding pipelines, semantic search**Databases:** Qdrant, Milvus, MongoDB**Tools:** Git, Bash, GitHub, GitHub Copilot, Postman, Swagger / OpenAPI**Concepts:** SDLC, OOP, Algorithm Design, Microservices, RESTful APIs, OAuth2 Authentication, Legacy Migration

EDUCATION

Manipal University Jaipur

Jaipur, India

Bachelor of Technology in Computer Science

Jun 2022

PROJECTS

IntelliSell — WhatsApp AI Sales Chatbot

- Engineered an AI-powered WhatsApp chatbot using Python and FastAPI, automating customer engagement with personalized offers, real-time responses, and order updates.
- Built a dynamic knowledge base by parsing and chunking PDF content, stored in Milvus Vector Database via Zilliz Cloud (53% conversational accuracy lift); automated onboarding and messaging workflows using Puppeteer.